



Scalability & Future Plan

Date	28 September 2026
Team ID	SWTID-2026-4103
Project Name	AI FAQ Assistant API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state, ensuring long-term viability and impact.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited support for very large numbers of users and concurrent API requests	May affect response time when user traffic increases	High
2	AI-generated FAQ content depends on external Google Gemini AI services	AI functionality may be affected if the external service is unavailable	Medium
3	Current system mainly focuses on FAQ creation, search, and AI-assisted support	Additional advanced customer-support features may be required in the future	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports authenticated and public users through REST APIs	Introduce load balancing and horizontal scaling of Node.js servers to support more concurrent users
2	Data Storage	MongoDB is used for storing users, FAQs, and related data	Scale MongoDB storage and optimize database indexing for larger FAQ datasets
3	Performance	Mongoose indexing and RESTful APIs provide efficient data access	Add caching, optimized queries, and asynchronous processing to improve response time
4	Security	JWT authentication, bcrypt password hashing, authorization middleware, and schema validation are implemented	Strengthen security with advanced access controls, monitoring, rate limiting, and regular security updates

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Advanced AI-powered FAQ generation and improved semantic search	3–6 Months	Faster and more accurate customer query resolution

Phase 3	Scalable infrastructure, caching, load balancing, and enhanced database optimization	6–12 Months	Improved performance and support for larger user loads
Phase 4	Scalable infrastructure, caching, load balancing, and enhanced database optimization	12–18 Months	Expanded functionality and improved long-term customer support automation